Visual Correspondence Hallucination

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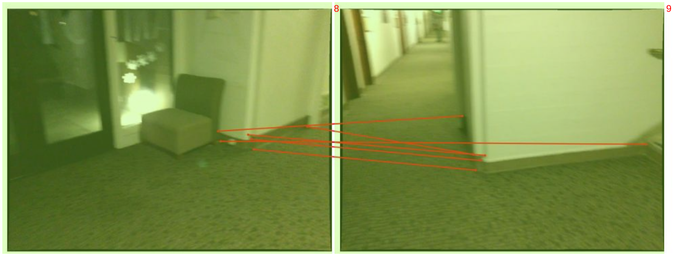
ICLR

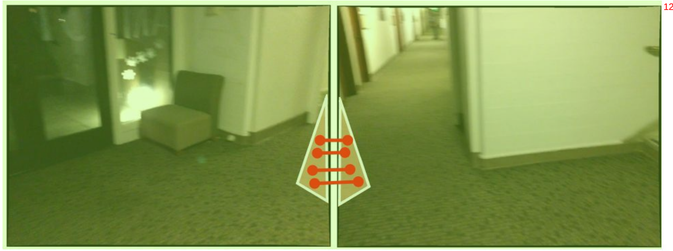
1. Low-overlap keypoint matching

- Keypoint matching is key for fundamental matrix & camera pose estimation
- In low-overlap scenarios however, finding correspondences is very challenging
- Existing methods will tend to find correspondences in non-covisible areas

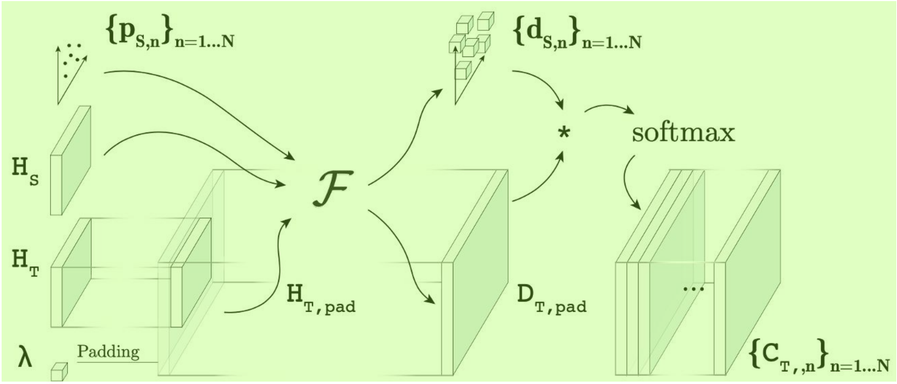


SuperPoint + SuperGlue

- Correct matches within image boundaries provide poor constraints



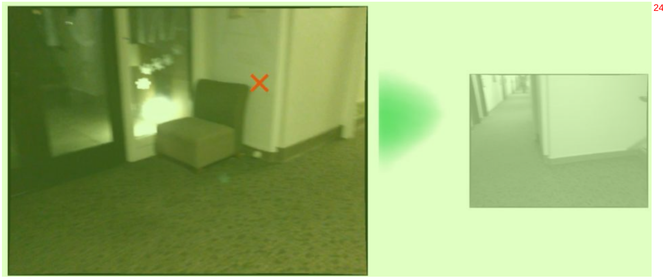
2. Introducing NeurHal



- We predict correspondence maps on extended field-of-views
- Training is done on both covisible and non-covisible samples
- We resort to a Transformer-based image processing backbone

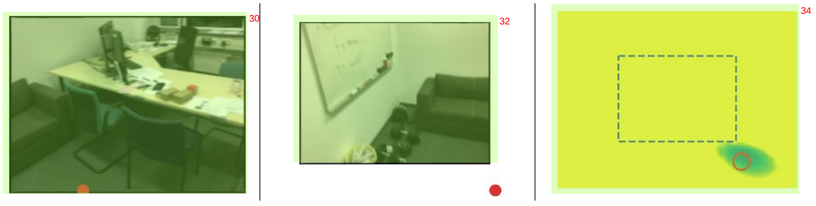
3. Application to absolute camera pose estimation

- We provide quantitative analysis of NeurHal's ability to hallucinate



NeurHal on validation image pair (novel scene)

- NeurHal achieves sota on absolute pose estimation for low-overlap image pairs
- More results in our paper !



Source Target Predicted NRE Map